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METHOD IN MOTION

THINK LIKE A PROGRAMMER

VOLUME 2: HOW COMPUTERS THINK



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Understanding the Systems, Logic, and Technology Behind Modern
Computing

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METHOD IN MOTION

WELCOME FUTURE PROGRAMMER

Programming is more than just writing code. Computers use systems, memory, logic, and communication to complete tasks and solve problems every day. In this volume, future programmers will explore how computers think, how programs work together, and how technology powers apps, websites, artificial intelligence, and modern digital systems.

INSIDE THIS VOLUME

// Inputs & Outputs	// Networks
// Data	// AI
// Binary	// Automation
// Memory	// APIs
// Functions	// Web Pages

Understanding technology begins with understanding systems.



INPUTS AND OUTPUTS

WHAT IS IT?

Computers take in information called inputs and respond with outputs. An input can be a click, tap, voice command, or typed message. The output is what the computer does after receiving the information. Understanding inputs and outputs helps kids see how humans interact with technology every day.

KEY TERMS

Input, Output, Response, Interaction

REAL LIFE USE

Pressing a button on a remote control changes the TV channel.

TRY THIS

Name one input and output you used today.

QUICK TIP



Computers listen for inputs before giving outputs.



DATA

WHAT IS IT?

Data is information that computers collect and use. Data can include numbers, words, pictures, sounds, or videos. Computers organize data so it can be searched, stored, and used later. Learning about data helps children understand how apps and websites remember information.

KEY TERMS

Data, Information,
Store, Collect

REAL LIFE USE

A music app saves your favorite songs as data.

TRY THIS

Think of 3 kinds of data a computer can store.

QUICK TIP



Data helps computers remember information.



BINARY

WHAT IS IT?

Binary is the language computers use to process information. Instead of letters and words, computers use combinations of 1s and 0s. These numbers help computers understand instructions and store information. Learning binary introduces kids to how computers communicate internally.

KEY TERMS

Binary, 1s and 0s, Code, Signals

REAL LIFE USE

A computer game uses binary to load graphics and sounds.

TRY THIS

Try writing a pattern using only 1s and 0s.

QUICK TIP



Computers use binary to understand instructions.



MEMORY

WHAT IS IT?

Computer memory stores information so programs can use it later. Memory helps devices remember apps, pictures, games, and instructions. Some memory is temporary while programs are running, and some stays saved for later use. Understanding memory helps children learn how devices keep information organized.

KEY TERMS

Memory, Storage, Save, Information

REAL LIFE USE

Saving a photo on a phone uses memory storage.

TRY THIS

Name something your device remembers for you.

QUICK TIP



Memory helps computers keep track of information.



FUNCTIONS

WHAT IS IT?

A function is a group of instructions designed to complete a specific task. Coders use functions to organize programs and avoid repeating the same code many times. Functions help make programs cleaner and easier to understand.

KEY TERMS

Function, Task, Instructions, Organize

REAL LIFE USE

A calculator app uses functions to solve math problems.

TRY THIS

Think of a task that repeats often every day.

QUICK TIP



Functions help programmers save time.



EVENTS

WHAT IS IT?

Events are actions that make something happen in a program. Clicking a button, pressing a key, or touching a screen can trigger an event. Programs wait for events before responding with actions.

KEY TERMS

Event, Trigger, Action, Response

REAL LIFE USE

Pressing play on a video starts an event.

TRY THIS

Name an event that happens on your favorite app.

QUICK TIP



Events help programs know when to respond.



USER INTERFACES

WHAT IS IT?

A user interface is the part of a program people interact with. Buttons, menus, icons, and screens are all parts of an interface. Good interfaces help users understand how to use technology easily.

KEY TERMS

Interface, Screen, Button, User

REAL LIFE USE

A game controller is part of a user interface.

TRY THIS

Look around your device and name 3 interface elements.

QUICK TIP



Interfaces help humans use technology.



NETWORKS

WHAT IS IT?

Networks connect computers and devices so they can share information. The internet is one large network connecting people around the world. Networks allow messages, videos, games, and websites to work together.

KEY TERMS

Network, Internet, Connection, Share

REAL LIFE USE

Sending a text message uses a network connection.

TRY THIS

Name one thing you use the internet for.

QUICK TIP



Networks help devices communicate.



ARTIFICIAL INTELLIGENCE

WHAT IS IT?

Artificial intelligence helps computers learn patterns and make decisions. AI can recognize voices, recommend videos, and answer questions. AI systems use large amounts of data to improve over time.

KEY TERMS

Artificial Intelligence,
Learn, Data, Decisions

REAL LIFE USE

A streaming app
suggesting movies uses
AI.

TRY THIS

Think of one device
that might use AI.

QUICK TIP



AI helps computers solve complex
problems.



AUTOMATION

WHAT IS IT?

Automation happens when technology completes tasks automatically. Coders create automation to save time and reduce repeated work. Automation can control lights, send reminders, or organize information.

KEY TERMS

Automation,
Automatic, Task,
Process

REAL LIFE USE

A robot vacuum cleaning by itself is automation.

TRY THIS

Name one task you wish could happen automatically.

QUICK TIP



Automation helps make work faster.



ENCRYPTION

WHAT IS IT?

Encryption protects information by turning it into secret code. Computers use encryption to keep passwords and messages safe online. Only the correct system or person can unlock encrypted information.

KEY TERMS

Encryption, Security, Secret, Password

REAL LIFE USE

Online banking uses encryption to protect accounts.

TRY THIS

Think of something important that should stay private.

QUICK TIP



Encryption helps keep digital information safe.



APIS

WHAT IS IT?

APIs allow programs and apps to communicate with each other. APIs help different systems share information and work together smoothly. Many websites and apps use APIs every day behind the scenes.

KEY TERMS

API, Connect, Share, System

REAL LIFE USE

A weather app using online weather information uses APIs.

TRY THIS

Think of two apps that might work together.

QUICK TIP



APIs help technology communicate.



GAME LOGIC

WHAT IS IT?

Game logic controls the rules and actions inside a video game. It tells the game what happens when players move, score points, or complete challenges. Coders use logic to create fair and fun gameplay.

KEY TERMS

Game Logic, Rules, Actions, Score

REAL LIFE USE

Collecting coins in a game uses programmed game logic.

TRY THIS

Think about one rule from your favorite game.

QUICK TIP



Game logic controls how games work.



WEB PAGES

WHAT IS IT?

Web pages are digital pages people view using browsers on the internet. Coders build web pages using programming languages and design tools. Websites can include text, pictures, videos, and buttons.

KEY TERMS

Web Page, Browser, Website, Internet

REAL LIFE USE

Visiting a school website means opening a web page.

TRY THIS

Name one website you visit often.

QUICK TIP



Web pages are built using code.



PROGRAMS WORKING TOGETHER

WHAT IS IT?

Many apps and programs work together to complete tasks. One program may send information while another receives and uses it. Modern technology depends on systems communicating smoothly with each other.

KEY TERMS

Program, System, Share, Connect

REAL LIFE USE

Ordering food online uses several connected programs at once.

TRY THIS

Think of one app that connects with another app.

QUICK TIP



Modern technology depends on connected systems.



VOLUME 2 SUMMARY

HOW COMPUTERS THINK

In this volume, you explored how modern technology works behind the scenes. You learned how websites, apps, servers, databases, cloud systems, and digital tools help people communicate, create, play, and solve problems every day.

You also discovered how coding powers interactive experiences like games, virtual reality, augmented reality, robots, and future technology. These concepts help build the foundation for understanding how computers think and how programmers create digital systems.

Technology is constantly evolving, and every coder starts by learning the systems behind it.



KEY DEFINITIONS



HTML — The structure of a web page.



CSS — The design and style of a website.



JavaScript — Adds interaction and movement to websites.



Apps — Software programs designed for specific tasks.



Websites — Connected web pages found online.



Servers — Computers that deliver online information.



Cloud Computing — Storing files and programs online.



Databases — Organized collections of digital information.



Game Design — Creating rules, worlds, and gameplay experiences.



Cyber Safety — Staying safe while using technology online.



Robots — Machines programmed to complete tasks automatically.



Digital Art — Creative artwork made using technology.



Augmented Reality (AR) — Digital elements added to the real world.



Virtual Reality (VR) — Immersive digital environments.



Future Technology — New inventions and ideas shaping the future.



“Technology becomes powerful when creativity and logic work together.””